

STATS 8 Midterm 2 (Discussion Worksheet Answers)

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1. The Youth Risk Behavior Survey (YRBS) is a survey conducted by the Centers for Disease Control and Prevention (CDC) in the spring once every two years, to learn about health-related risky behaviors in 9th-12th graders in the U.S. In 2021, when asked whether they had been in a physical fight in the 12 months prior to the survey, 18.34% of the students surveyed said that they had. 51.31% of students surveyed identified as male. Of those who identified as male, 23.24% said they had been in a physical fight in the 12 months prior to the survey.

Let event A be the event that a randomly selected student from this survey identified as male.
Let event B be the event that a randomly selected student from this survey had been in a physical fight in the 12 months prior to the survey.

- (a) What does 0.2324, or 23.24%, represent? Use the probability notation we learned in class.

$P(B | A)$

- (b) What is $P(A \text{ and } B)$? Round your answer to 2 decimal places.

Solution: $P(A \text{ and } B) = P(A)P(B | A)$

$$P(A \text{ and } B) = (0.5131)(0.2324) \approx 0.12$$

- (c) What is $P(A \text{ or } B)$? Round your answer to 2 decimal places.

Solution: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$$P(A \text{ or } B) = 0.5131 + 0.1834 - 0.1192 \approx 0.58$$

- (d) In plain English, what quantity does $P(A | B^c)$ represent?

Solution: This is the probability that a randomly selected student is male given that they had not been in a physical fight in the prior 12 months.

(e) What is $P(A \mid B^c)$?

Solution:

$$\begin{aligned} P(A \mid B^c) &= \frac{P(A \text{ and } B^c)}{P(B^c)} \text{ (conditional probability definition)} \\ &= \frac{P(A \text{ and } B^c)}{1 - P(B)} \text{ (complement rule)} \\ &= \frac{P(A) - P(A \text{ and } B)}{1 - P(B)} \text{ (law of total probability)} \\ &= \frac{0.5131 - 0.1192}{1 - 0.1834} \approx 0.48 \end{aligned}$$

(f) Are the events A and B independent? Why or why not?

Solution: Events A and B are independent if $P(B \mid A) = P(B)$. We are given $P(B \mid A) = 0.2324$ and $P(B) = 0.1834$, so the events are not independent.